LAB 7: Collaboration Diagram – Library Management System

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Aim

To draw the collaboration (communication) diagram of the Library Management System for the Issue Book scenario, showing how the objects are linked and which messages they exchange.

Theory

A collaboration diagram, called a communication diagram in UML 2, is an interaction diagram that shows the objects of a system, the links between them and the messages that are passed along those links. It shows the same information as a sequence diagram, but the stress is on the structure (who is connected to whom) and not on the time order.

The elements used in the diagram are:

- **Object / Actor:** a participant of the interaction, drawn as a box.
- **Link:** a solid line between two objects that can communicate with each other.
- **Message:** an arrow or label written along a link. The direction of the message is shown by the arrow symbol.
- **Sequence number:** the number written before every message (1, 2, 3 ...) that shows the order of the messages.
- **Guard condition:** a condition in square brackets, such as [valid], written before a message. The message is sent only if the condition is true.

Scenario

The scenario drawn is Issue Book, which is the same scenario as the sequence diagram of Lab 5. The librarian sends the member ID and the ISBN to the system. The system checks the member and the book. If the request is valid, it creates the issue record, reduces the stock of the book and confirms to the librarian. Otherwise the request is rejected.

Objects Used

Object	Role
Librarian	The actor who starts the issue request.
:LibrarySystem	The control object that receives the request and coordinates the other objects.

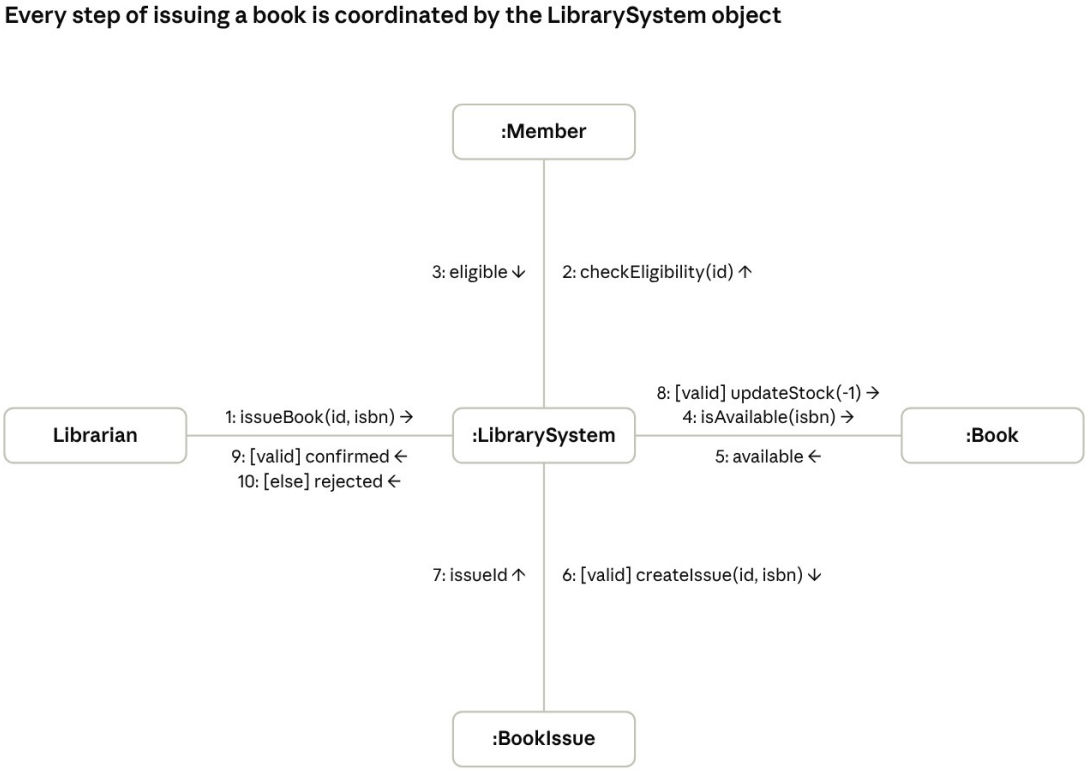
:Member	Holds the details of the member and tells whether the member is eligible.
Object	Role
:Book	Holds the details of the book and its available copies.
:BookIssue	The record that is created when a book is issued.

Messages

No.	From	To	Message	Description
1	Librarian	:LibrarySystem	issueBook(id, isbn)	The librarian sends the member ID and the book ISBN.
2	:LibrarySystem	:Member	checkEligibility(id)	The system checks the pending fine and the issue limit of the member.
3	:Member	:LibrarySystem	eligible	The eligibility result is returned.
4	:LibrarySystem	:Book	isAvailable(isbn)	The system checks whether a copy of the book is available.
5	:Book	:LibrarySystem	available	The availability result is returned.
6	:LibrarySystem	:BookIssue	[valid] createIssue(id, isbn)	If the request is valid, the issue record is created.
7	:BookIssue	:LibrarySystem	issueId	The ID of the new issue record is returned.
8	:LibrarySystem	:Book	[valid] updateStock(-1)	If the request is valid, the available copies are reduced by one.
9	:LibrarySystem	Librarian	[valid] confirmed	If the request is valid, the confirmation is shown to the librarian.
10	:LibrarySystem	Librarian	[else] rejected	Otherwise the request is rejected.

Collaboration Diagram

The collaboration diagram of the Issue Book scenario is shown below. The arrow symbols next to each message show its direction along the link.



collaboration diagram · 5 objects, 10 numbered messages

Figure 1: Collaboration diagram of the Issue Book scenario in the Library Management System.

Sequence Diagram and Collaboration Diagram

Point	Sequence diagram	Collaboration diagram
Main stress	Time order of the messages	Links between the objects
Order of messages	Shown by the vertical position	Shown by the sequence numbers
Layout	Objects in a row at the top with lifelines	Objects placed freely with links between them
Best used for	Showing the flow of one scenario in detail	Showing which objects work together

Conclusion

The collaboration diagram of the Issue Book scenario of the Library Management System was drawn. It shows the Librarian, LibrarySystem, Member, Book and BookIssue objects, the links between them and the numbered messages that they exchange.